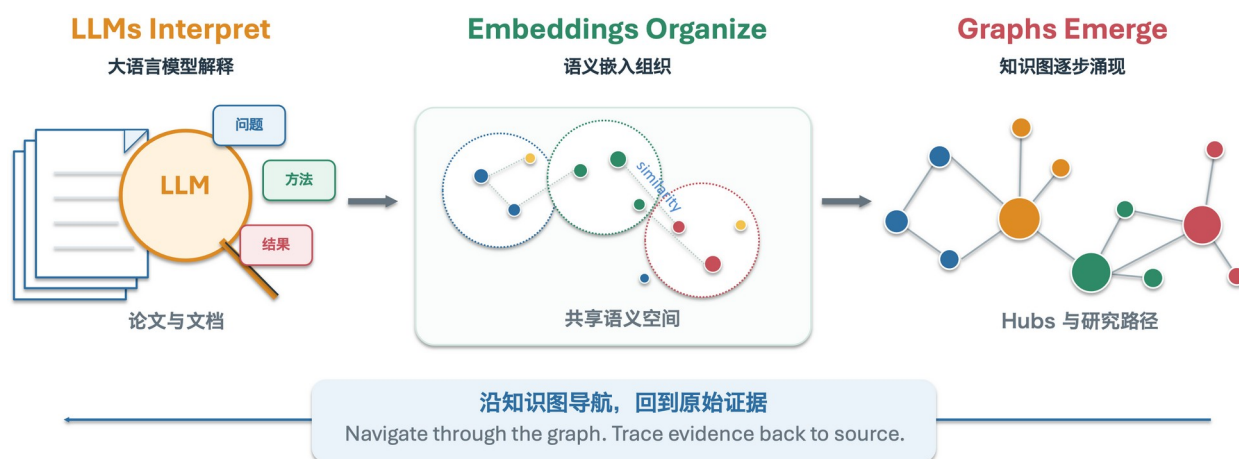


## AGENT-DRIVEN SCIENTIFIC KNOWLEDGE SYSTEM

## ASKS □ □ □ □ □ □

## I

## LLMs Interpret, Embeddings Organize, Graphs Emerge



ASKS 让每次局部理解进入一个可持续、可导航、可追溯的知识状态。

ASKS Agent-Driven Scientific Knowledge System

□ □ □ □

arXiv [arXiv:2608.29612](https://arxiv.org/abs/2608.29612)

[github.com/ranshiju/Ran-ASKS](https://github.com/ranshiju/Ran-ASKS)

| 2026 9

$$[1]$$

AGENTS.md

```

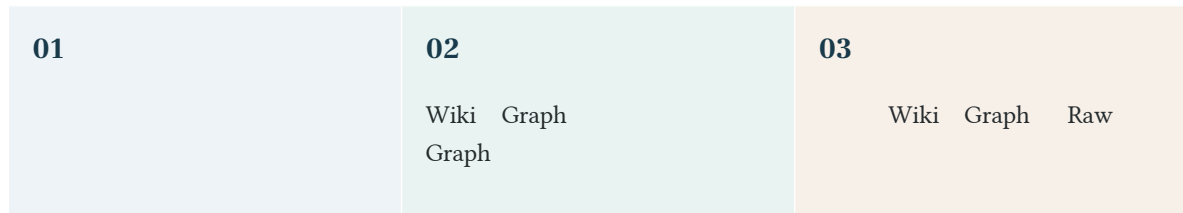
graph LR
    Embedding --- LLM
    LLM --- API1[API]
    API1 --- API2[API]
    API2 --- Env[.env]
    Embedding --- Env
    LLM --- Env
    Embedding --- API1
    LLM --- API2
    Embedding --- Env
  
```

AI

## SKS

[2] ASKS

Agent



ASKS

Wiki Graph

Agent

## 1 LLMs Interpret, Embeddings Organize, Graphs Emerge

ASKS

| LLM       |  |  | Wiki |
|-----------|--|--|------|
| Embedding |  |  |      |
| Graph     |  |  |      |

### LLMs Interpret

Wiki

LLM

### Embeddings Organize

Embedding

[4] ASKS

Embedding

Embedding

Hub

Agent

## Graphs Emerge

ASKS

Hub

“ ”

Hub

## 2 Raw Wiki Graph

ASKS

FAIR PROV-DM RO-Crate [5-7]

| <b>Raw</b>   |     |       |  |
|--------------|-----|-------|--|
| <b>Wiki</b>  |     |       |  |
| <b>Graph</b> | Hub | Agent |  |

## Wiki

Wiki

Raw Wiki

Wiki

Semantic MediaWiki[8]

## Graph

Graph

Hub

Agent

[3] ASKS Graph

|     |      |       |       |
|-----|------|-------|-------|
| Raw | Wiki | Graph | Graph |
|-----|------|-------|-------|

Wiki

Graph

Raw Wiki Graph

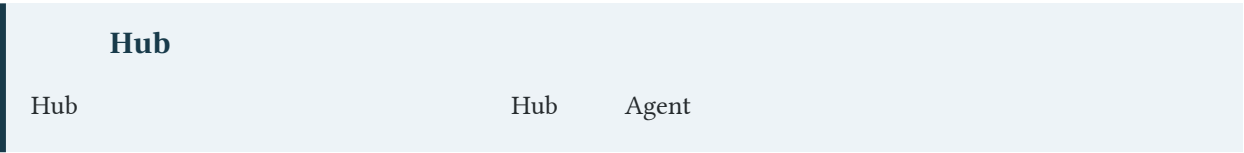
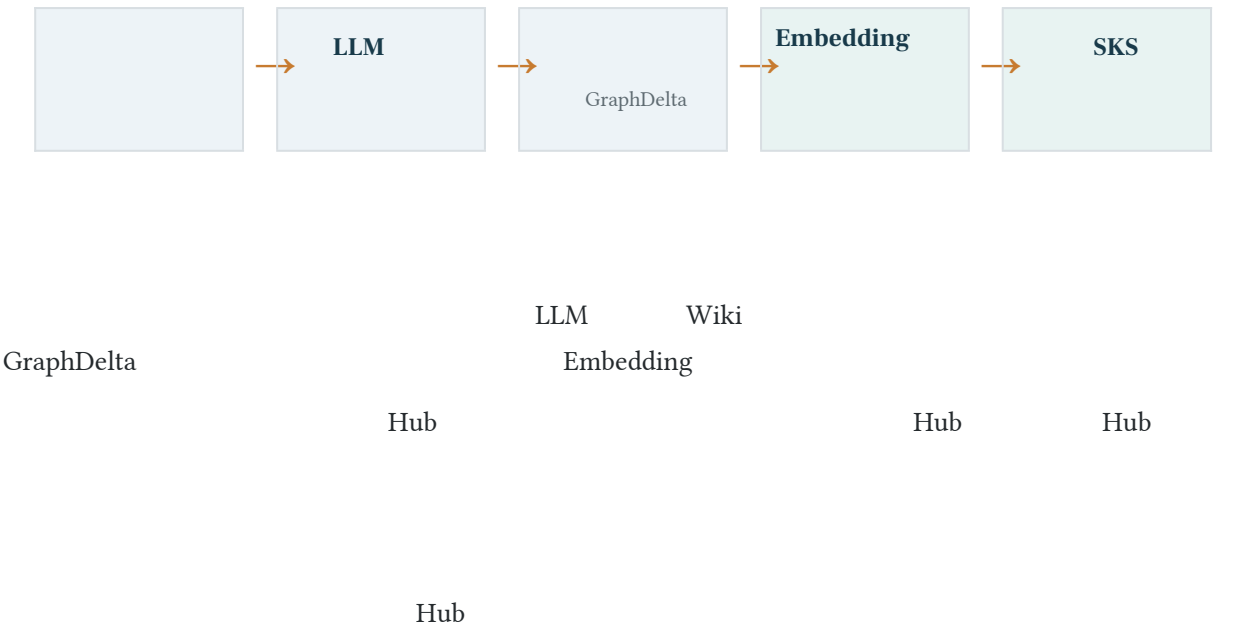
Graph

ASKS

Graph

## 3 Graph

ASKS



4

ASKS

|      | Wiki Graph |
|------|------------|
|      | LLM        |
|      |            |
|      |            |
|      | Embedding  |
|      | Hub        |
| gent | Agent      |

ASKS

operations/engineering/graph.yaml

Agent

engineering\_graph.py

impact

Markdown

YAML

Python

Agent

Agent

ASKS

Agent

Agent

Agent

DSH

Agent

DSH

ASKS

Python Agent

ToolRegistry

hook

guard

SessionLog

AgentLoop

DSH

DeepSeek Harness

ASKS

Agent

DSH

Raw Wiki

graph.db

5 56

ASKS

56

18

Hub

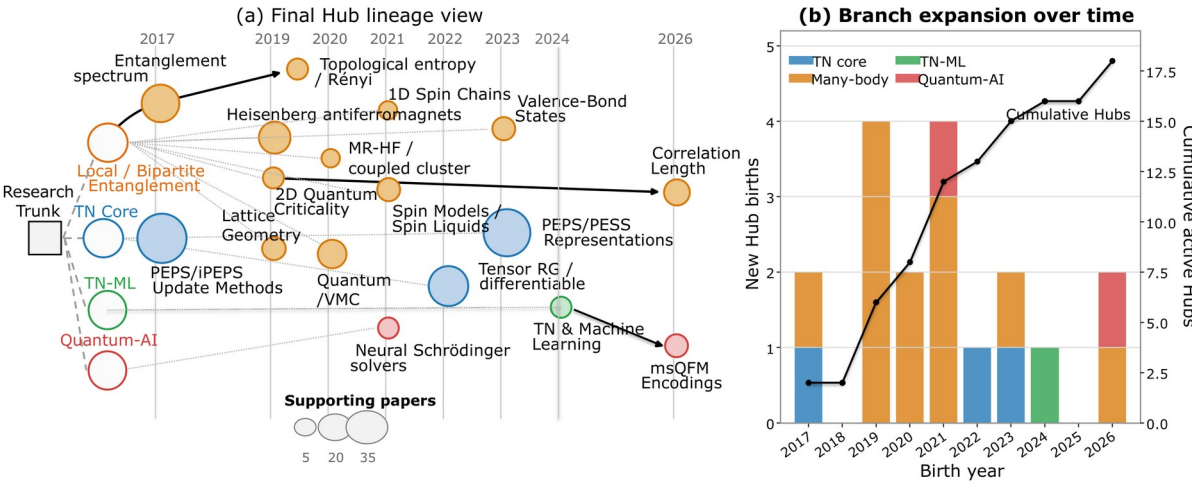
50

Hub 6

Hub

0.47%

Hub



1 56

AI

Hub

APS Physics Subject Headings  
Hub

18 Hub Hub 0.774 0.5

Hub Hub

56 Embedding Graph

18 21 Hub Hub 0.379 0.669  
67.9% 83.9% 89.3%

ASKS Graph

|  |                      |     |
|--|----------------------|-----|
|  |                      |     |
|  | 5618Hub 500.47%      |     |
|  | 0.5PhySHHub0.774     | Hub |
|  | 1821HubHub0.3790.669 |     |

Hub

ASKS

Graph

ASKS

Agent

Graph

Wiki

Graph

Agent

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ASKS

|                                  |  | SKS                                                       |
|----------------------------------|--|-----------------------------------------------------------|
| <a href="#">MinerU</a>           |  | PDF 56<br>MinerU                                          |
| <a href="#">DeepSeek Harness</a> |  | Everything is a Plugin hook<br>guard DSH Python<br>Cordis |
| <a href="#">Semantica</a>        |  | ASKS                                                      |
| <a href="#">Docling</a>          |  | MinerU                                                    |

|                             |  | SKS  |
|-----------------------------|--|------|
| <a href="#">PyMuPDF</a>     |  | PDF  |
| <a href="#">python-pptx</a> |  | PPTX |

[GitHub](#) | [arXiv](#)

ASKS

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ASKS v0.2.1

ASKS

1.1.0

Agent-Driven Scientific Knowledge System

arXiv v1

arXiv:2608.29612

arXiv v1

2026 9 1

Ran-ASKS v0.2.0

Ran-ASKS v0.2.2

1.0.0

Ran-